

SUBJECT: Strategic Data Architecture: Deploying a "Dual-Layer" Hybrid Model for Enterprise HR Analytics

1. Executive Summary

To support our goal of predicting employee attrition at an MNC scale (50,000+ employees, daily transactional changes), we must move beyond traditional data modeling. This report proposes a "Dual-Layer Hybrid Architecture." This new approach uniquely bridges the gap between human-readable analytics (Power BI) and machine-learning requirements (Databricks), ensuring we maintain a single, historically accurate source of truth without duplicating workflows.

2. Initial Data Optimization

Before applying any architecture, we optimized the dataset. Out of the 35 raw HR attributes, three are static constants (EmployeeCount, StandardHours, Over18). At an enterprise scale, processing useless columns in Databricks wastes compute and storage. We have isolated these, leaving 32 highly actionable attributes to be architected.

3. The Problem with Traditional "Either/Or" Ways

Historically, organizing data forced us to choose between two competing needs, both of which fail at our required scale:

- **The "Human" Way (Analytics/Star Schema):** Optimizes data for Power BI. It groups data perfectly and tracks historical changes (e.g., tracking an employee's promotion history). *The Flaw:* AI models in Databricks cannot read this format; they require mathematical translations (numbers, not text categories).
- **The "AI" Way (Flat ML Clustering):** Translates all 32 attributes into math (e.g., converting "Male/Female" to 1s and 0s) so an algorithm can read it. *The Flaw:* It flattens history. If an employee changes roles, the old data is overwritten. For 50,000+ employees with daily movement, the AI loses the ability to learn from past states.

We can no longer afford to build separate databases for Power BI and AI.

4. The Solution: The "Dual-Layer" Hybrid Way (Deep Dive)

To solve this, we are not choosing a traditional way; we are combining the "Human" way and the "AI" way into a single, elegant architecture.

Think of it like a universal translator sitting on top of a secure filing cabinet. The filing cabinet holds the truth; the translator adapts it for whoever is asking.

How we group the 32 attributes:

Instead of grouping by strict database rules, we group the 32 attributes by Logical Themes that make sense to both an HR executive and an AI pattern-recognition algorithm:

- **Identity Theme:** Age, Gender, Marital Status, Education.
- **Environment Theme:** Department, Job Role, Distance from home, Overtime.
- **Compensation Theme:** Monthly Income, Salary Hike, Stock Options.
- **Sentiment Theme:** Environment Satisfaction, Relationship Satisfaction, Work-Life Balance.
- **Tenure Theme:** Years at company, Years in current role, Promotions.

The Two Layers in Action:

- **Layer 1: The Foundation (Built for Power BI & History)**

We physically store the 32 attributes in Databricks using the Theme groups above. Because it is structured this way, if an employee is promoted from Sales to Manager, the system creates a new record with a timestamp, preserving the history. Power BI connects directly here. It is fast, compressed, and human-readable.

- **Layer 2: The Translation View (Built for the AI Model)**

We build a virtual "View" in Databricks that sits on top of Layer 1. It doesn't store new data. Instead, it automatically reads Layer 1 and applies the math the AI needs in real-time (e.g., translating the "Environment Theme" text into scaled decimal numbers). The ML model connects here.

Why this Hybrid Way is the breakthrough:

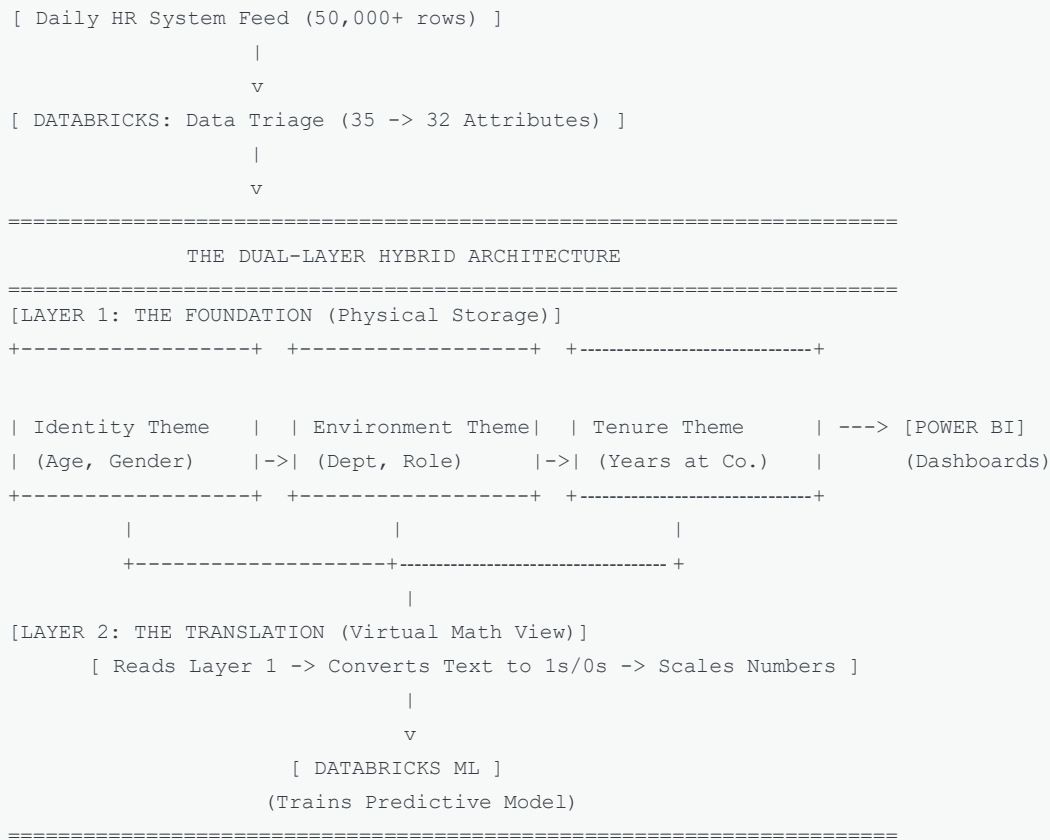
It eliminates the "AI vs. Dashboard" war. When 500 new employees are hired and 50 separate tomorrow, the data is entered once into Layer 1. Power BI immediately sees the updated dashboards, and the AI model immediately sees the newly translated math. Zero data duplication, zero sync errors.

5. Tech Stack Synergy (Databricks & Power BI)

This new architecture unlocks the full potential of our existing stack:

- **Databricks:** Acts as the engine for both layers. We use Delta Lake for Layer 1 (ensuring absolute data reliability and historical tracking). We use Databricks SQL for Layer 2 (performing the AI translations on the fly).
- **Power BI:** Connects strictly to Layer 1. Because Layer 1 is grouped by clean "Themes" and doesn't contain messy AI-translation math, Power BI's engine compresses it flawlessly, resulting in sub-second dashboard loading for HR leadership.

6. Architecture Workflow Diagram



7. Project Roadmap: From System to Decision

By adopting this Hybrid architecture, we can seamlessly execute the DIKW (Data-Information-Knowledge-Wisdom) roadmap for the business:

- **Phase 1: Enterprise System -> Data (Weeks 1-2)**

Action: Ingest raw data into Databricks. Build Layer 1 (The Foundation) grouped by the 5 Logical Themes.

Result: A single, historically tracking source of truth for 50,000+ employees.

- **Phase 2: Data -> Metrics & Information (Weeks 3-4)**

Action: Connect Power BI to Layer 1. Build Layer 2 (The Translation Views).

Result: HR gets fast, theme-based dashboards (Information). The AI gets mathematically scaled data (Metrics) to begin learning.

- **Phase 3: Information -> Knowledge (Weeks 5-6)**

Action: Train the Machine Learning model on Layer 2. The AI correlates the "Compensation Theme" with the "Sentiment Theme" to find hidden drivers of attrition.

Result: The system moves from showing what is happening to understanding why it is happening (Knowledge).

- **Phase 4: Knowledge -> Decision (Ongoing)**

Action: Automate the pipeline. The model generates a daily "Flight Risk Score" for every employee, pushed back to Power BI.

Result: MNC Leadership moves from reactive reporting to proactive decision-making (e.g., intervening with a high-performing employee before they accept a competitor's offer).

8. Conclusion

The "Dual-Layer" Hybrid way is not just a technical upgrade; it is a strategic necessity. By fusing human-centric analytics and machine-centric math into one pipeline, we ensure that as our workforce grows and changes daily, our data architecture remains fast, historically accurate, and intelligently predictive.